

ECE 447

Lesson 08

Amplitude

Modulation, Part 1



UNITED STATES
AIR FORCE
ACADEMY



Life/Leadership Lesson of the Day

■ Proofread everything!

- Emails
- Letters
- Memos
- Text messages
- ...

■ and signs?



SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **HW problems...** or problems with HW?
 - **Lab 2 Assignment.** The assignment associated with Lab 2 is due Lesson 10 (Tue, 1 Sep) by 2359 via Gradescope upload.
- When are you going to get things graded??? Working on it...

DID YOU DO THE READING?

What was the video *baseband* for the old analog TV signals?

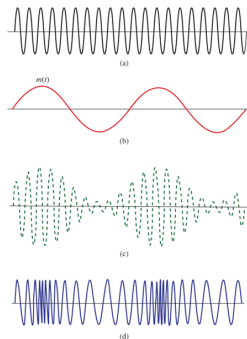
Where do we see *baseband communications*?

Where do we see *carrier communications*?

Modulation means *changing* something about a signal - and not always just shifting the signal's frequencies

MODULATION

- Message signal: $m(t)$
- General carrier modulated signal:
$$s(t) = A(t) \cos [2\pi f_c t + \phi(t)]$$
- Amplitude (linear) modulation, $A(t)$
 - $s_1(t) = m(t) \cdot \cos (2\pi f_c t)$
 - $S_1(f) = \frac{1}{2}M(f - f_c) + \frac{1}{2}M(f + f_c)$
- Angular modulation, f_c or $\phi(t)$
 - Message modulates frequency or phase
 - Notice the general equation doesn't have $f_c(t)$ - frequency modulation is a little different - more in future!



DOUBLE-SIDEBAND, SUPPRESSED-CARRIER

(DSB-SC) Simplest form of AM

- Learned this in ECE 215
- $\varphi_{DSB-SC}(t) = A_c m(t) \cos(\omega_c t)$
- Over-, under-, or 100%-modulated?
- Demodulation
 - Same as modulation with LPF
 - What if $f_c < B$?
 - Reality, $f_c/B \gg 1$ to avoid distortion
 - Difficult to achieve synchronized carrier signals for receiver

